Solving for a Variable in a Formula

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Formula

A **formula** is a mathematical rule that shows the relationship between quantities using numbers, variables, and operations.

$$A = lw$$

$$\text{Area} = \text{length} \times \text{width}$$

Parts of a Formula

$$A = lw$$

A	area
l	length
w	width

Each letter is a **variable**.

Solving for a Variable

To **solve a variable in terms of other variables** means to isolate one variable on one side of the equation while expressing it using the other variables.

$$A = lw \quad \Rightarrow \quad w = \frac{A}{l}$$

Properties of Equality

Use the **properties of equality** to perform the same operation on both sides of an equation.

Keep the equation balanced.

$$x + 3 = 10$$
$$x + 3 - 3 = 10 - 3$$
$$x = 7$$

Inverse Operations

Apply the **inverse operations** to isolate the desired variable.

Operation		**Inverse**
Addition	$\leftrightarrow$	Subtraction
Multiplication	$\leftrightarrow$	Division
Squaring	$\leftrightarrow$	Square root

Memory Tip

Undo in reverse order.

1. Undo addition and subtraction first.
2. Then undo multiplication and division.

This is like **PEMDAS backwards**.

Variable in a Fraction

To isolate a variable that appears in a fraction, multiply both sides by the denominator or use equivalent algebraic steps.

$$s = \frac{d}{t}$$

$$s \cdot t = \frac{d}{t} \cdot t$$

$$d = st$$

Variable in Several Terms

To isolate a variable that appears in several terms:

1. Collect the terms containing that variable on one side.
2. Factor out the variable.
3. Divide by its coefficient.

One Formula, Many Forms

A formula may be rearranged to solve for different variables without changing the relationship it represents.

$$A = lw$$

$$l = \frac{A}{w} \qquad w = \frac{A}{l}$$

Examples

Example 1

Solve for h: $V = lwh$.

$$V = lwh$$

$$\frac{V}{lw} = h$$

$$h = \frac{V}{lw}$$

Example 2

Solve for c: $P = a + b + c$.

$$P = a + b + c$$

$$P - a - b = c$$

$$c = P - a - b$$

Example 3

Solve for x: $y = 2x + 5$.

$$y = 2x + 5$$
$$y - 5 = 2x$$
$$\frac{y - 5}{2} = x$$

Example 4

Solve for x: $ax + bx = c$.

$$ax + bx = c$$

$$x(a + b) = c$$

$$x = \frac{c}{a + b}$$

Example 5

Solve for s: $A = s^2$.

$$A = s^2$$
$$\sqrt{A} = s$$

Squaring and taking the square root are inverse operations.

Practice Exercises

Practice 1

Rearrange the formula to solve for w.

$$A = lw$$

Practice 2

Rearrange the formula to solve for l.

$$P = 2l + 2w$$

Practice 3

Rearrange the formula to solve for x.

$$y = 3x + b$$

Practice 4

Rearrange the formula to solve for h.

$$A = \frac{1}{2}bh$$

Practice 5

Rearrange the formula to solve for r.

$$C = 2\pi r$$

Practice 6

Rearrange the formula to solve for t.

$$d = vt$$

Practice 7

Rearrange the formula to solve for b.

$$y = mx + b$$

Practice 8

Rearrange the formula to solve for x.

$$ax + by = c$$

Practice 9

Rearrange the formula to solve for w.

$$P = 2l + 2w$$

Practice 10

Rearrange the formula to solve for x.

$$y = \frac{x + a}{b}$$

Activities

Activity 1

Rearrange the formula to solve for w.

$$A = sw$$

Activity 2

Rearrange the formula to solve for l.

$$P = 4l + 4w$$

Activity 3

Rearrange the formula to solve for x.

$$y = 5x + c$$

Activity 4

Rearrange the formula to solve for h.

$$A = \frac{1}{2}ch$$

Activity 5

Rearrange the formula to solve for r.

$$C = 4\pi r$$

Activity 6

Rearrange the formula to solve for t.

$$d = at$$

Activity 7

Rearrange the formula to solve for b.

$$y = nx + b$$

Activity 8

Rearrange the formula to solve for x.

$$px + qy = r$$

Activity 9

Rearrange the formula to solve for w.

$$P = 6l + 2w$$

Activity 10

Rearrange the formula to solve for x.

$$y = \frac{x + c}{d}$$

Thank you!